

Purna Sainath Reddy V

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AI/ML Engineer focused on building production-ready intelligent systems across AGI research, AI agents, graph machine learning, computer vision, MLOps and AI security. Built all 35+ projects @ **Portfolio: purnareddy.me** - Each of these projects exists for a reason: a real problem worth solving. Together they reflect my pursuit of depth in every domain, not just breadth across them.

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Mohan Babu University - B.Tech CSE (AI & ML)
Narayana College (MPC)

CGPA: 8.15
MARKS: 96%

2022 - 2026
2020 - 2022

SKILLS

- **AI Research & Specialised Systems:** Program Synthesis, ARC-AGI, DreamCoder/Wake-Sleep, RAG (FAISS, LangChain), VLMs, Diffusion/4D, Quantum ML (Qiskit, PennyLane), Whisper, ChromaDB, OpenAI API
- **Machine Learning & AI:** PyTorch, PyTorch Geometric, GraphSAGE, XGBoost, scikit-learn, YOLOv8, ONNX, NumPy, MLflow, SHAP, Multi-Agent Systems
- **Languages:** Python, SQL, Java, C++, TypeScript/JavaScript
- **Vision, 3D & Systems:** OpenCV, COLMAP (Photogrammetry), Edge AI (AMD Ryzen NPU), Digital Twin, Diffusion View-Synthesis, Deterministic Optimization, Three.js, WebGL
- **Data Science & Analytics:** Statistical Modeling, Regression, EDA, Drift Detection (PSI), Data Visualization (Power BI, Plotly, Recharts), Excel, Web Scraping
- **Web & Infrastructure:** React, Next.js, Node.js, FastAPI, Django, PostgreSQL, Neo4j, Tailwind, Docker, Databricks, Azure Arc, Privacy Engineering (k-Anonymity, Differential Privacy)

KEY PROJECTS

NexusGuard - GraphSAGE, Louvain, PyTorch Geometric, FastAPI

[live demo ↗](#)

- Built a graph-native identity risk-scoring engine (**GraphSAGE + Louvain**, PyTorch Geometric) that judges each login against its functional peer group rather than its own history: **0.993 AUC-ROC, 1.3% FPR (4× lower than ML-only)** at <15ms latency.
- Fused five signals (peer alignment, geo-velocity, temporal, device, entity) into a 0–100 risk score driving adaptive auth, validated on the CERT r6.2 benchmark with k-anonymity and differential privacy by design.

AbstractMind-Research - Program Synthesis, DreamCoder, ARC-AGI, NumPy

[GitHub ↗](#)

- Built a **self-improving program-synthesis engine for ARC-AGI** (DreamCoder-style wake-sleep, enumerative search and MDL compression in NumPy) that invents and reuses its own abstractions.
- It spontaneously discovered a cross-module abstraction reused 28×, beating a matched baseline **11.8% vs 10.5% across 400 tasks**; full results in the linked research paper.

VigilCore - Databricks, XGBoost, SHAP, MLflow, LLM Eval

[GitHub ↗](#)

- Built **VigilCore**, a **self-monitoring credit-risk MLOps pipeline on the Databricks Lakehouse**: a Bronze/Silver/Gold medallion architecture with an XGBoost + SHAP champion model in the MLflow registry and PSI-based drift detection.
- Automated champion/challenger retraining with triple-gate promotion and an **LLM-evaluation stage that catches hallucinations** in loan-rejection explanations via a 4-signal composite check.

SentinelOS - AI Agents, LLMs, Whisper STT, AppleScript, Voice

[GitHub ↗](#)

- Built **SentinelOS**, a **JARVIS-like AI operating-system layer for macOS** that sees the system and acts on it: file management, shell execution, app control via AppleScript and web access, all driven by an LLM backend (GitHub Copilot, 64K–128K context) through text and real-time voice.
- Engineered a fully local voice pipeline (Whisper STT + native TTS, no API) and a **Guardian security supervisor** that risk-reviews every action before execution, hard-blocking destructive operations.

AgentForge - Agentic AI, Next.js, React 19, Control Plane

[live demo ↗](#)

- Built **SAO**, an **enterprise control plane for autonomous AI agents** that hold write access to real systems: identity and intent governance, least-privilege scopes, behavioral-security boundaries, kill switches and a full audit trail (Next.js, React 19).
- Designed an orchestration canvas for visual multi-agent workflows with human approval gates, plus live mission-control monitoring with severity alerts and deviation history.

OTHER LIVE PROJECTS TO CHECK OUT

- **UrbanTwin** - YOLOv8, ONNX, COLMAP, AMD Ryzen AI NPU, Streamlit [live demo ↗](#)
Drone video into a 3D digital twin that auto-detects defects, on the edge.
- **CloudShift** - Azure Arc, React, Redux, Node/Express, SAP BTP [live demo ↗](#)
Hybrid-cloud migration with rollback, IoT carbon tracking and Zero Trust.
- **CampusHub** - Next.js 15, React 19, Neo4j, FastAPI, Turborepo [live demo ↗](#)
Graph-based platform rendering each student's journey as a live roadmap.
- **5G O-RAN Optimizer** - Python, FastAPI, D3.js, Docker [live demo ↗](#)
Deterministic 5G fronthaul optimizer; cuts link capacity by up to 88.5%.
- **Inventory Risk Intelligence** - scikit-learn (Decision Tree), FastAPI, PostgreSQL, React [live demo ↗](#)
Predicts SKU expiry risk (~95%) and recommends discounts, urgency and actions.
- **PathFinderAI Recommendation** - Next.js 15, React, TypeScript, OpenAI API [live demo ↗](#)
Turns any job title into a roadmap of skills, courses, projects and salary.
- **Internship Retrieval Platform** - React, TypeScript, FastAPI, Tailwind, SQLite [live demo ↗](#)
Internship finder with scam detection, credibility scoring and smart ranking.
- **AI Tutor** - React, Vite, OpenAI API [live demo ↗](#)
AI hints/solutions for students and AI grading for teachers, multi-language.
- **TicketFlow** - Django, DRF, PostgreSQL, scikit-learn [live demo ↗](#)
Jira-Lite support system with ML triage and TF-IDF duplicate detection.
- **HeritageGlobe** - Three.js, WebGL, JavaScript [live demo ↗](#)
Interactive 3D viewer of 25+ procedurally-built world monuments in 360°.
- **EstateLens** - Leaflet.js, Chart.js, Tailwind, JavaScript [live demo ↗](#)
Real-estate SPA with a map, per-listing trust report and liveability radar.

EXPERIENCE

Website Development Intern, Vidya - The Change

Apr 2025 – May 2025

- Built accessible, user-friendly web interfaces for digital access to learning resources.
- Optimized platform performance for low-connectivity users, with a non-profit team.

Finlatics Business Analytics Experience Program

Dec 2025 – Feb 2026

- Built Excel and Power BI dashboards tracking KPIs, sales patterns and profitability.
- Applied regression to find key pricing factors and model revenue impact.

RESEARCH & PUBLICATIONS

- **AGI Program Synthesis** - DreamCoder, Wake-Sleep, ARC-AGI, MDL Compression [read paper ↗](#)
First-authored "PrimaLearn Bootstrapping" paper; invents its own abstractions, 11.8% vs 10.5% baseline.
- **TemporalForge: 4D Completion** - Diffusion, Zero123++/SV3D, Optical Flow [GitHub ↗](#)
Turns monocular video into a complete, explorable 360° 4D scene with consistent generated views.
- **MotionProbe** - VLMs (CLIP), DRL, PyTorch, Distributed (DDP) [GitHub ↗](#)
A VLM critic scores generative-motion naturalness, fed back as a DRL perceptual reward.
- **Quantum Machine Learning** - Qiskit, PennyLane, VQC, Quantum Kernels [GitHub ↗](#)
IEEE-format paper where VQC and quantum-kernel models beat classical baselines.
- **CortexRAG** - RAG, Phi-3, llama.cpp, FAISS, LangChain [GitHub ↗](#)
Offline, citation-backed neuroscience RAG running fully on-device with zero API cost.

ACHIEVEMENTS

- **Built and deployed 15 full-stack & ML systems end-to-end** — from data pipelines and model training to live, production-grade web apps.
- **Self-directed depth across 7 technical domains**— AI security, ML/AGI research, computer vision, data science, full-stack, AI agents, and quantum.
- **Two first-authored AI research papers ready for publication** (ARC-AGI program synthesis; variational quantum machine learning).
- **13 live demos shipped** across a 35+ project portfolio spanning 7 technical domains.